

ADVANCED CALCULUS & DIFFERENTIAL EQUATIONS

Examination Year: 2026 | Maximum Marks: 100 | Total Questions: 4 | Time Allowed: 3 Hours

GENERAL INSTRUCTIONS:

1. All questions are compulsory unless specified otherwise.
2. Candidates must show all working, intermediate steps, and derivations to earn full marks.
3. Draw diagrams wherever necessary; label all parts clearly.
4. Figures, graphs and diagrams are provided with the respective question.
5. Use of scientific calculator is permitted unless otherwise notified.

Q.No	1	2	3	4	Total
Marks	25	30	25	20	100

Question 1

[25 Marks]

Find the directional derivative of the function $f(x, y) = x^2 y + \sin(xy)$ at the point $(1, \pi)$ in the direction of the vector $\mathbf{v} = (3, 4)$. Normalize the direction vector before computation.

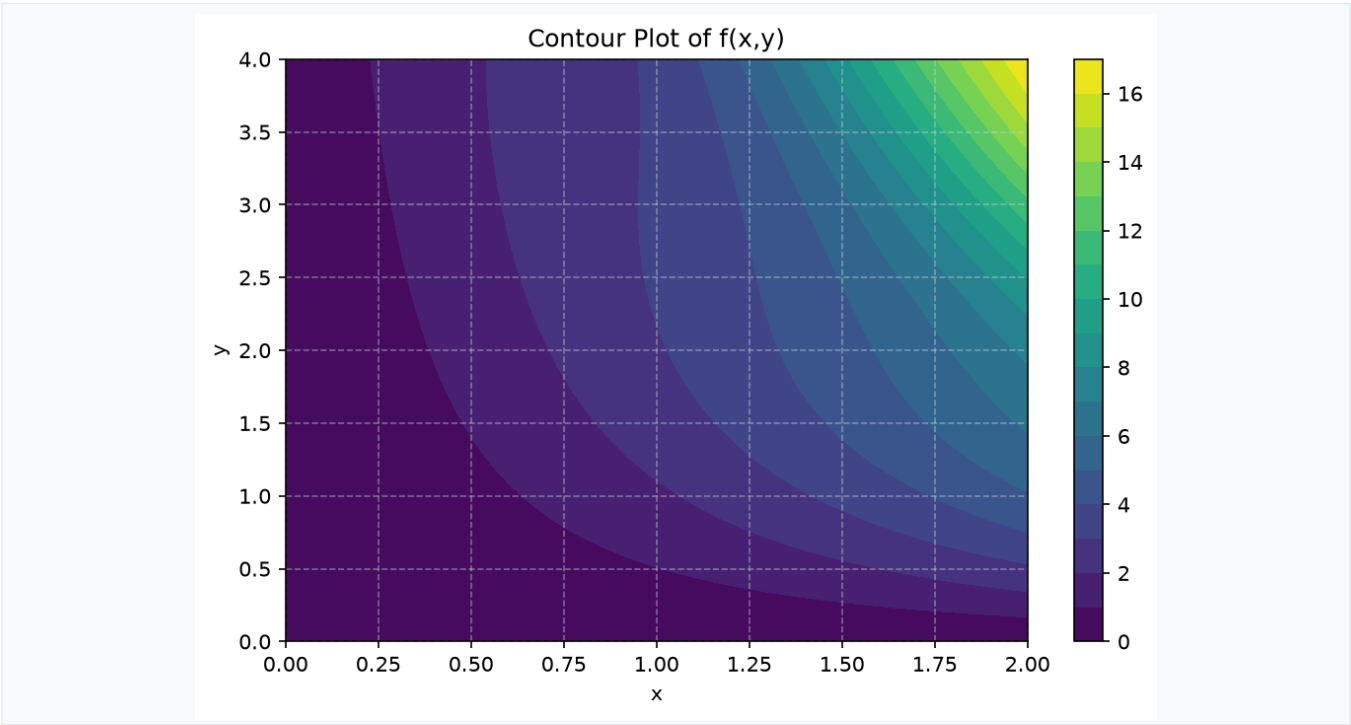


Figure 1

Question 2

[30 Marks]

Compute the surface integral $\int_S (x^2 + y^2) \, dS$ where S is the part of the paraboloid $z = 4 - x^2 - y^2$ that lies above the xy -plane. Use appropriate parameterization and show all steps.

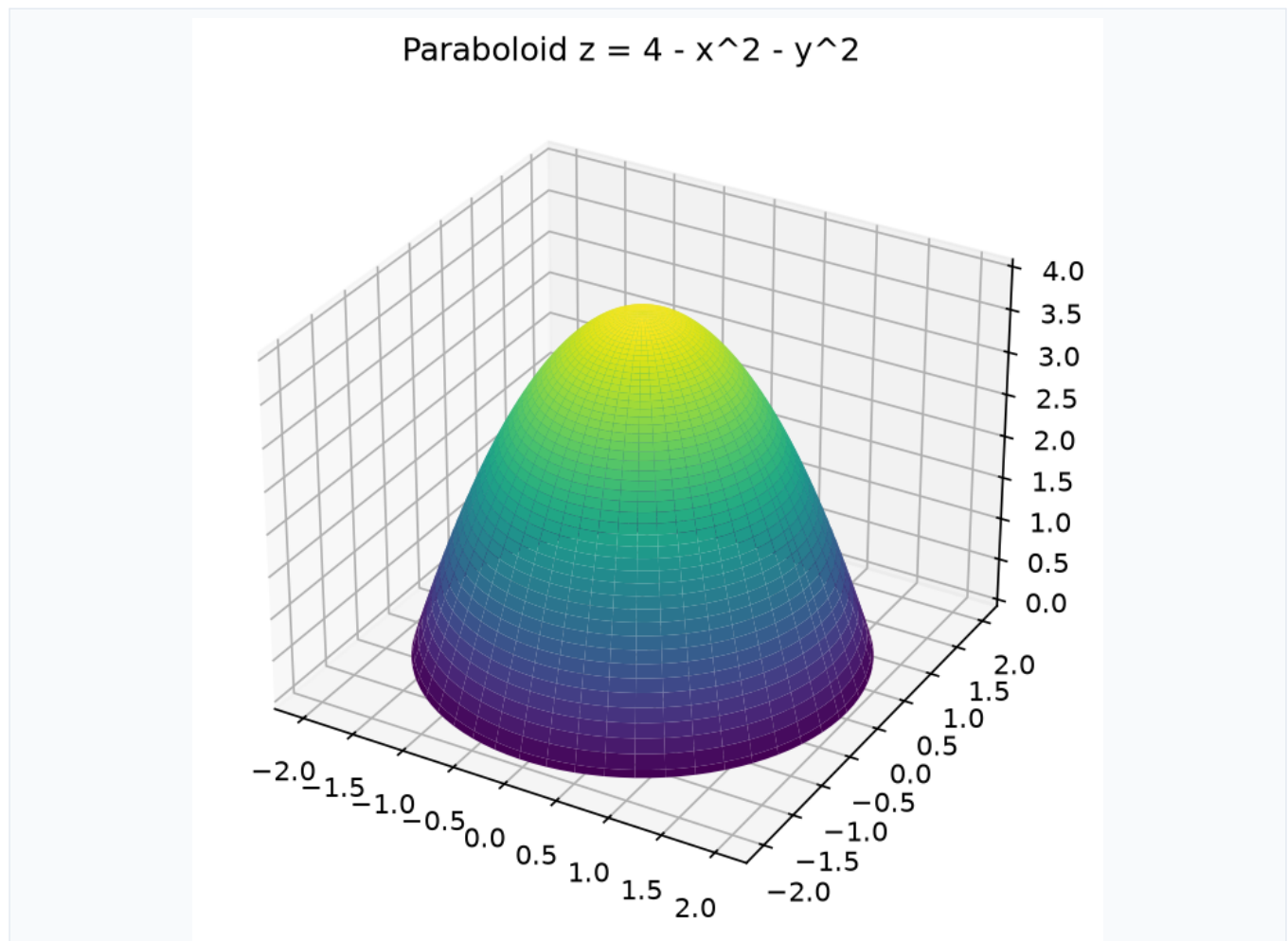


Figure 2

Question 3**[25 Marks]**

Solve the first-order differential equation $(dy / dx) + y \tan x = \sin 2x$ with the initial condition $y(0) = 1$. Show all steps and verify your solution.

Question 4**[20 Marks]**

Which of the following is the general solution to the differential equation $x^2 (d^2y / dx^2) - 2x (dy / dx) + 2y = 0$?

- (a) $y = C_1 x + C_2 x^2$
- (b) $y = C_1 x + C_2 e^x$
- (c) $y = C_1 x^2 + C_2 x^3$
- (d) $y = C_1 x^{-1} + C_2 x^2$

Justify your answer by solving the differential equation.
